WORKED SOLUTION

STEP III 2005, Question 7

Pure • Integration by Substitution

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s3-q7

The key idea

The opening result says that a factor of x^m hidden inside an integrand can be treated as a single new variable $u = x^m$, as long as the integrand also has a lone factor of $\frac{1}{x}$ outside. So for each integral we pull out a factor of x from the denominator, read off which power x^m is left and what the function f is, and do the much easier u-integral instead.

Solution

The general result

We are told that $\mathrm{F}'(u) = \dfrac{1}{u\,\mathrm{f}(u)}$. Differentiating $\mathrm{F}(x^m)$ by the chain rule (for $x > 0$, so that x^m makes sense for any real m),

$$\frac{\mathrm{d}}{\mathrm{d}x}\mathrm{F}(x^m) = \mathrm{F}'(x^m)\cdot mx^{m-1} = \frac{mx^{m-1}}{x^m\,\mathrm{f}(x^m)} = \frac{m}{x\,\mathrm{f}(x^m)}.$$

So $\mathrm{F}(x^m)$ is an antiderivative of $\dfrac{m}{x\,\mathrm{f}(x^m)}$, which is exactly the result required:

$$\int \frac{m}{x\,\mathrm{f}(x^m)}\,\mathrm{d}x = \mathrm{F}(x^m) + c.$$

Equivalently, this is the substitution $u = x^m$, $\mathrm{d}u = mx^{m-1}\,\mathrm{d}x$, read backwards. The condition $m \neq 0$ matters in what follows: we shall divide by m.

In practice we use it as

$$\int \frac{1}{x\,\mathrm{f}(x^m)}\,\mathrm{d}x = \frac{1}{m}\,\mathrm{F}(x^m) + c. \qquad (*)$$

Part (i)

Taking out a factor of x,

$$\frac{1}{x^n - x} = \frac{1}{x\,(x^{n-1} - 1)}.$$

This has the form in $(*)$ with $m = n - 1$ and $\mathrm{f}(u) = u - 1$. We need $n \neq 1$; when $n = 1$ the denominator $x^n - x$ is identically zero and there is no integral to find.

Now we need F, an antiderivative of $\dfrac{1}{u(u-1)}$. Partial fractions give

$$\frac{1}{u(u-1)} = \frac{1}{u-1} - \frac{1}{u}, \qquad \text{so} \qquad \mathrm{F}(u) = \ln|u-1| - \ln|u| = \ln\left|\frac{u-1}{u}\right|.$$

Putting $u = x^{n-1}$ into $(*)$:

$$\boxed{\int \frac{1}{x^n - x}\,\mathrm{d}x = \frac{1}{n-1}\ln\left|\frac{x^{n-1}-1}{x^{n-1}}\right| + c, \qquad n \neq 1.}$$

The modulus signs let this hold on any interval where the integrand is defined (for instance on $0 < x < 1$ as well as $x > 1$; and for $x < 0$ when n is an integer).

Part (ii)

Here we must be a little careful taking x out of the square root. We work with $x > 0$, where x^n is defined for every n and $\sqrt{x^2} = x$. Then

$$\sqrt{x^n + x^2} = \sqrt{x^2\,(x^{n-2} + 1)} = x\sqrt{1 + x^{n-2}},$$

so the integrand is

$$\frac{1}{x\sqrt{1 + x^{n-2}}},$$

which has the form in $(*)$ with $m = n - 2$ and $\mathrm{f}(u) = \sqrt{1+u}$.

? **Ask yourself** What happens when $n = 2$?

Then $m = 0$ and the general result does not apply. But the integral is easy directly: the integrand is $\dfrac{1}{\sqrt{2x^2}} = \dfrac{1}{\sqrt{2}\,x}$, so

$$\int \frac{1}{\sqrt{x^2 + x^2}}\,\mathrm{d}x = \frac{1}{\sqrt{2}}\ln x + c \qquad (n = 2).$$

For $n \neq 2$ we need $\mathrm{F}(u) = \displaystyle\int \frac{1}{u\sqrt{1+u}}\,\mathrm{d}u$, where $u = x^{n-2} > 0$. Substitute $t = \sqrt{1+u}$, so $u = t^2 - 1$, $\mathrm{d}u = 2t\,\mathrm{d}t$ and $t > 1$:

$$\int \frac{1}{u\sqrt{1+u}}\,\mathrm{d}u = \int \frac{2t}{(t^2-1)\,t}\,\mathrm{d}t = \int \left(\frac{1}{t-1} - \frac{1}{t+1}\right)\mathrm{d}t = \ln\left(\frac{t-1}{t+1}\right),$$

with no modulus needed since $t > 1$. This tidies up nicely: multiplying top and bottom by $t-1$ and using $t^2 - 1 = u$,

$$\frac{t-1}{t+1} = \frac{(t-1)^2}{t^2-1} = \frac{\left(\sqrt{1+u}-1\right)^2}{u}, \qquad \text{so} \qquad \mathrm{F}(u) = 2\ln\left(\sqrt{1+u}-1\right) - \ln u.$$

Now put $u = x^{n-2}$ into $(*)$ with $m = n-2$. The term $-\ln u$ becomes $-(n-2)\ln x$, and dividing by $m = n-2$ turns it into $-\ln x$:

$$\boxed{\int \frac{1}{\sqrt{x^n + x^2}}\,\mathrm{d}x = \frac{2}{n-2}\ln\left(\sqrt{1+x^{n-2}}-1\right) - \ln x + c, \qquad n \neq 2,\ x > 0.}$$

Equivalently, the answer is $\dfrac{1}{n-2}\ln\left(\dfrac{\sqrt{1+x^{n-2}}-1}{\sqrt{1+x^{n-2}}+1}\right) + c$; either form is fine.

Checks

- Differentiate the answers back. For (i) with $n = 2$ the formula gives $\ln|x-1| - \ln|x|$, whose derivative is $\frac{1}{x-1} - \frac{1}{x} = \frac{1}{x^2-x}$, as it should be.
- For (i) with $n = 3$, the formula gives $\displaystyle\int_2^3 \frac{\mathrm{d}x}{x^3 - x} = \tfrac{1}{2}\ln\tfrac{32}{27} \approx 0.084950$, matching Simpson's rule.
- For (ii) with $n = 3$, the formula gives $\displaystyle\int_1^3 \frac{\mathrm{d}x}{\sqrt{x^3 + x^2}} = \ln\frac{3+2\sqrt{2}}{3} \approx 0.664135$, again matching numerical integration. We also tested both formulae for many values of n, including negative and non-integer ones.

Common slips

- Forgetting the factor $\frac{1}{m}$: the general result has m in the numerator, and our integrands don't, so we must divide by m.
- Writing $\sqrt{x^2} = x$ without comment. It is true only for $x \geq 0$; for $x < 0$ it is $-x$ (or $|x|$).
- Missing the exceptional values $n = 1$ in (i) and $n = 2$ in (ii), where $m = 0$.

Beyond STEP

The opening result is the substitution rule seen from the other side: substitution is just the chain rule run backwards, and that is how it is proved in a first analysis course, with careful hypotheses on the functions involved. Integrals such as (ii), involving square roots of polynomials, lead to the question of which integrals can be done in elementary functions at all; $\int \mathrm{d}x/\sqrt{1+x^4}$, for instance, cannot. For a careful treatment

of the fundamental theorem of calculus and substitution, see Abbott, *Understanding Analysis*.

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